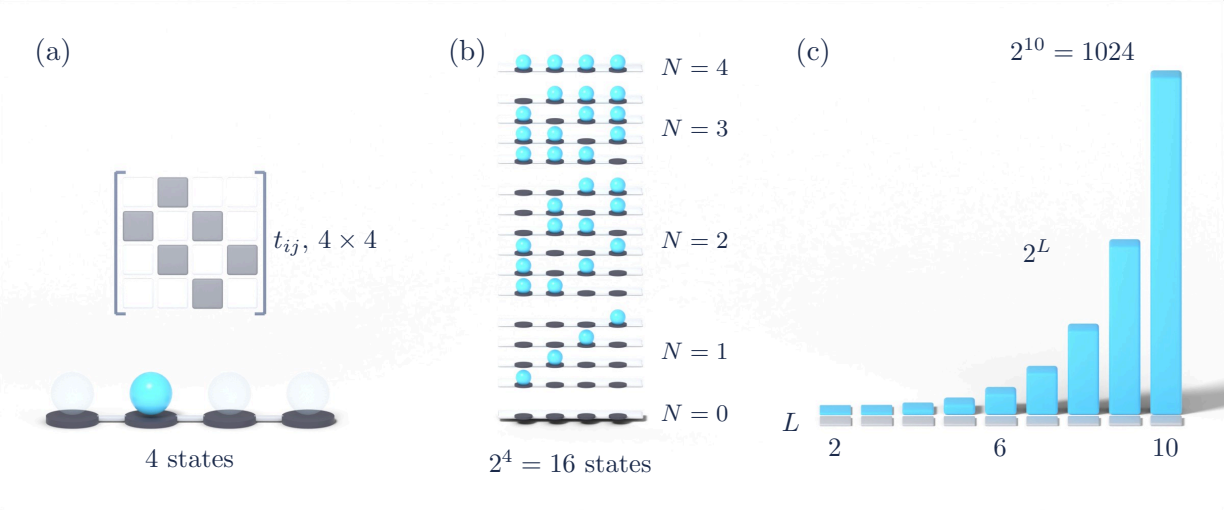

(a)
t_{ij}, 4×4
4 states
(b)
$N = 4$
$N = 3$
$N = 2$
$N = 1$
$N = 0$
$2^4 = 16$ states
(c)
$2^{10} = 1024$
2^L
L
2
6
10